

Solution to B3 continued

P1

$$A = \begin{pmatrix} 0 & 0 & 2 & 4 \\ 1 & 5 & 2 & 1 \\ 1 & 5 & 3 & 3 \end{pmatrix} \xrightarrow{\text{Row 2} - \text{Row 3}} \begin{pmatrix} 0 & 0 & 2 & 4 \\ 0 & 0 & -1 & -2 \\ 1 & 5 & 3 & 3 \end{pmatrix}$$

$$\xrightarrow{\text{Row 2} + \frac{1}{2} \text{Row 1}} \begin{pmatrix} 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \\ 1 & 5 & 3 & 3 \end{pmatrix} \xrightarrow{\text{Row 2} \leftrightarrow \text{Row 3}}$$

$$\begin{pmatrix} 0 & 0 & 2 & 4 \\ 1 & 5 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{Row 1} \leftrightarrow \text{Row 2}} \begin{pmatrix} 1 & 5 & 3 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\frac{1}{2} \times \text{Row 2}}$$

$$\begin{pmatrix} 1 & 5 & 3 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{Row 1} - 3 \cdot \text{Row 2}} \begin{pmatrix} 1 & 5 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

which is the desired RREF of A.

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B4. Write your solution to B4 below.

P2Consider the same set of operations on $(A|\underline{b})$ as in P1 &:

$$\left(\begin{array}{cccc|c} 0 & 0 & 2 & 4 & 2 \\ 1 & 5 & 2 & 1 & 4 \\ 1 & 5 & 3 & 3 & 5 \end{array} \right) \xrightarrow{\text{Row 2} - \text{Row 3}} \left(\begin{array}{cccc|c} 0 & 0 & 2 & 4 & 2 \\ 0 & 0 & -1 & -2 & -1 \\ 1 & 5 & 3 & 3 & 5 \end{array} \right)$$

$$\xrightarrow{\text{Row 2} \leftarrow \frac{1}{2} \text{Row 2}} \left(\begin{array}{cccc|c} 0 & 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 5 & 3 & 3 & 5 \end{array} \right) \xrightarrow{\begin{array}{l} \text{Row 2} \leftrightarrow \text{Row 3} \\ \text{Row 1} \leftrightarrow \text{Row 2} \end{array}}$$

$$\left(\begin{array}{cccc|c} 1 & 5 & 3 & 3 & 5 \\ 0 & 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{\frac{1}{2} \times \text{Row 2}} \left(\begin{array}{cccc|c} 1 & 5 & 3 & 3 & 5 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$\xrightarrow{\text{Row 1} - 3 \cdot \text{Row 2}} \left(\begin{array}{cccc|c} 1 & 5 & 0 & 3 & 2 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Because elementary row operations preserve the solutions space, $A\underline{x} = \underline{b}$ is equivalent to the following system of linear equations:

$$\left(\begin{array}{cccc} 1 & 5 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow x_3 + 2x_4 = 1 \Rightarrow x_3 = 1 - 2x_4 \quad (i)$$

$$x_1 + 2x_2 + 3x_4 = 2 \Rightarrow x_1 = 2 - 2x_2 - 3x_4$$

Solution to B2 continued

So that our solution set is:

$$\underline{x} = x_2 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -3 \\ 0 \\ -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad x_2, x_4 \in \mathbb{R}$$

↳ free variables.

Note that this is precisely all the solutions to $A\underline{x} = \underline{b}$: if it is easy to verify that there are indeed solutions, and if there is some $(x_1, x_2, x_3, x_4)^T$ which is not such a solution, it would violate the system of equations in (i).

Consider the subspace $W = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ -2 \\ 1 \end{pmatrix} \right\}$. We have already shown in class that $\text{span}(S)$ is indeed a subspace*. Then, using $\underline{p} = \begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, we have that

our solution set is $W + \underline{p}$, as for

$$\underline{x}_0 = \underbrace{\left\{ x_2^0 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4^0 \begin{pmatrix} -3 \\ 0 \\ -2 \\ 1 \end{pmatrix} \right\}}_{\in W} + \underbrace{\begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{\underline{p}}$$

a solution to $A\underline{x} = \underline{b}$, the first two terms ~~are~~ are in ~~span~~ W , so that $\underline{x}_0 = \underline{y}_0 + \underline{p}$ where $\underline{y}_0 \in W$.

Solution to B2 continued in pages 14 to 28? Write the page number here _____

* (If my memory is
* unless my memory is poorer than I already suspected.

B3. Write your solution to B3 below.

Let us convert $(A | \underline{b})$ to $(A_{REF} | \underline{b}_{REF})$ for convenience in formulating this:

$$\begin{pmatrix} 0 & 0 & 2 & 4 & | & b_1 \\ 1 & 5 & 2 & 1 & | & b_2 \\ 1 & 5 & 3 & 3 & | & b_3 \end{pmatrix} \xrightarrow[\text{Row 2} \leftarrow \frac{1}{2} \text{Row 1}]{\text{Row 2} - \text{Row 3}} \begin{pmatrix} 0 & 0 & 2 & 4 & | & b_1 \\ 0 & 0 & 0 & 0 & | & b_2 - b_3 + \frac{1}{2} b_1 \\ 1 & 5 & 3 & 3 & | & b_3 \end{pmatrix}$$

$$\xrightarrow[\text{Row 1} \leftrightarrow \text{Row 2}]{\text{Row 3} \leftrightarrow \text{Row 2}} \begin{pmatrix} 1 & 5 & 3 & 3 & | & b_3 \\ 0 & 0 & 2 & 4 & | & b_1 \\ 0 & 0 & 0 & 0 & | & b_2 - b_3 + \frac{1}{2} b_1 \end{pmatrix} \xrightarrow{\frac{1}{2} \text{Row 2}}$$

$$\begin{pmatrix} 1 & 5 & 3 & 3 & | & b_3 \\ 0 & 0 & 1 & 2 & | & \frac{1}{2} b_1 \\ 0 & 0 & 0 & 0 & | & b_2 - b_3 + \frac{1}{2} b_1 \end{pmatrix} \quad \text{As we have already shown}$$

in class, this has a solution iff there is no pivot in the last column. Clearly, then, $A\underline{x} = \underline{b}$ (equivalent system as row operations don't change solution set), has a solution iff

$$b_2 - b_3 + \frac{1}{2} b_1 = 0$$

□

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P4

Consider $M = \begin{pmatrix} 1 & 12 & 0 & -15 \\ 0 & 14 & -1 & -23 \end{pmatrix}$

so that

$$M \underline{x} = \underline{0} \Rightarrow \begin{cases} x_1 + 12x_2 - 15x_4 = 0 \\ 14x_2 - x_3 - 23x_4 = 0 \end{cases} \quad \text{and}$$

$$\Rightarrow \begin{cases} x_1 = -12x_2 + 15x_4 \\ 14x_2 - x_3 - 23x_4 = 0 \end{cases}$$

$$\Rightarrow \begin{aligned} x_1 &= -12x_2 + 15x_4 \\ x_3 &= 14x_2 - 23x_4 \end{aligned}$$

Hence, for $\underline{x} \neq \underline{0}$, $M \underline{x} = \underline{0}$,

$$\underline{x} = x_2 \begin{pmatrix} -12 \\ 1 \\ 14 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 15 \\ 0 \\ -23 \\ 1 \end{pmatrix}$$

where $x_2, x_4 \in \mathbb{R}$. Thus, the null space of M is precisely u .

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(a) and (b)

PS

§ (a) Consider the map $\underline{x} \mapsto A\underline{x}$, $f: \mathbb{R}^q \rightarrow \mathbb{R}^p$.
we note each vector, $A\underline{x} \in \mathbb{R}^p$ can be written as:

$$A\underline{x} = x_1 \underline{c}_1 + \dots + x_q \underline{c}_q$$

where $\underline{c}_i$ are the column vectors of A , by just matrix multiplication of $\underline{x}$ with A .

We note in particular that this implies the map

$\underline{x} \mapsto A\underline{x}$ is injective iff $\{\underline{c}_i\}$ the column vectors are linearly independent, if $\exists \underline{x}_1, \underline{x}_2 : A\underline{x}_1 = A\underline{x}_2$

$$\Rightarrow x_1 \underline{c}_1 + \dots + x_q \underline{c}_q = x'_1 \underline{c}_1 + \dots + x'_q \underline{c}_q$$

$$\Rightarrow \sum (x_i - x'_i) \underline{c}_i = 0 \Rightarrow x_i = x'_i \quad \forall i$$

Thus, $\underline{x}_1 = \underline{x}_2$. Similarly, if the column vectors were not independent, so that $\exists$ not all zero x_i :

$$x_1 \underline{c}_1 + \dots + x_q \underline{c}_q = 0$$

$\Rightarrow A\underline{x} = 0$ has solutions $(0, \dots, 0)^T$ and $(x_1, \dots, x_q)^T$ where $(0, \dots, 0)^T \neq (x_1, \dots, x_q)^T$, so that f would not be injective.

be injective.

$\underline{x} \mapsto A\underline{x}$: injective $\iff \underline{c}_i$ are lin. independent

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(being injective)

Note in particular that $A\underline{x} = \underline{b}$ having a unique solution is equivalent to the ~~rows of A being~~ all columns of A having a pivot, as we showed in class.

This is equivalent to the RREF of A being of the form:

$$\left(\begin{array}{cccc|c} 1 & 0 & \dots & 0 & \\ 0 & 1 & \dots & 0 & \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & \dots & 0 & \frac{1}{0} \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & \dots & \dots & 0 & \end{array} \right) \begin{array}{l} q \\ p \end{array}$$

Rem: $p \geq q$, as

each col. has a pivot

$\Rightarrow \exists$ row that has a pivot in that column

But the first q rows of A_{RREF} are simply $e_1, \dots, e_q$, which form a basis of $\mathbb{R}^q$. Therefore,

$\underline{r}_1^T, \dots, \underline{r}_p^T$ span $\mathbb{R}^q$, where $\underline{r}_i^T$ is the row vector of

these A_{RREF} .

Now, note that A_{RREF} is achieved by elem. row operations which are invertible, hence, and are linear combinations of the rows, i.e. we replace $\underline{r}_j^T$ with $c_{1j}\underline{r}_1^T + c_{2j}\underline{r}_2^T + \dots + c_{pj}\underline{r}_p^T$ s.t. $c_{jj} \neq 0$ (we may ignore swapping as we are not concerned with indices. ~~if you swap, then the condition replaces with some $\underline{r}_k^T$ while considering the span~~).

One weight is not zero because we can't scale by zero, and row. operations produce lin. combination as we can combine the scaling and adding by a multiple of a row to achieve a lin. combination repeatedly to get a lin. combination.

~~The fact that lin. combinations are~~

And clearly elem. row operations are themselves lin. combinations

$$a_i r_i \Leftrightarrow r_i' = a_i r_i + 0 r_2 + \dots + 0 r_n, \quad r_i + a r_j \Leftrightarrow r_i' = r_i + a r_j + 0 \dots$$

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Thus, the original row vectors of A are:

$$c_{11} \underline{r}_1^T + \dots + c_{p1} \underline{r}_p^T, \dots, c_{1p} \underline{r}_1^T + \dots + c_{pp} \underline{r}_p^T$$

where $\underline{r}_i^T$ are the row vectors of A_{RREF} , $c_{ii} \neq 0 \forall i$.

Therefore But this also ~~forms~~ spans $\mathbb{R}^n$ as for any $\underline{x}$ in $\mathbb{R}^n$, since $\{\underline{r}_i^T\}$ span it,

$$\underline{x} = a_1 \underline{r}_1^T + \dots + a_p \underline{r}_p^T$$

But then,

$$\underline{x} = \frac{a_1}{c_{11}} \left(\frac{a_{11} c_{11}}{c_{11}} + \frac{a_{21}}{c_{21}} + \dots + \frac{a_{p1}}{c_{p1}} \right) \underline{r}_1^T + \dots + \frac{a_p}{c_{pp}} \left(\frac{a_{1p} c_{1p}}{c_{1p}} + \frac{a_{2p}}{c_{2p}} + \dots + \frac{a_{pp} c_{pp}}{c_{pp}} \right) \underline{r}_p^T$$

Therefore, each of $\underline{r}_i^T$ are linear combinations of $\underline{r}_1^T, \dots, \underline{r}_p^T$.
we claim that $\underline{r}_1^T, \dots, \underline{r}_p^T$ spans $\mathbb{R}^n$.

Pf: Suppose not. Then,

$\exists \underline{s} \in \mathbb{R}^n$ s.t. $\underline{s}$ is not a linear combination of $\underline{r}_1^T, \dots, \underline{r}_p^T$. But we know that $\underline{s}$ is a linear combination of $\underline{r}_1^T, \dots, \underline{r}_p^T$, therefore

$$\underline{s} = a_1 \underline{r}_1^T + \dots + a_p \underline{r}_p^T. \quad 18$$

Since $\underline{r}_i^T = a_{i1} \underline{r}_1^T + \dots + a_{ip} \underline{r}_p^T$, $\underline{s}$ must also be a lin. combination of $\{\underline{r}_i^T\}$, contradicting our hypothesis

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Note that this statement also runs the opp. direction.
 if $\underline{r}_1^T, \dots, \underline{r}_p^T$ spans $\mathbb{R}^q$, by exactly the same proof, so
 must $\underline{r}_1'^T, \dots, \underline{r}_p'^T$ because row. operations are invertible implies
 that $\underline{r}_i'^T$ are linear combinations of $\underline{r}_1'^T, \dots, \underline{r}_p'^T$.

Hence we have shown a system of equivalences:

* $\underline{c}_1, \dots, \underline{c}_q$ ~~are~~ are lin. independent

$\Leftrightarrow \underline{x} \mapsto A\underline{x}$ is injective

$\Leftrightarrow$ each column of A_{REF} has a pivot

$\Leftrightarrow$ row vectors of A_{REF} span $\mathbb{R}^q$

[shown in class]

$\Leftrightarrow$ row vectors of A_{REF} span $\mathbb{R}^q$

Thus,

$\underline{c}_1, \dots, \underline{c}_q$ are lin. independent

$\Leftrightarrow$ row vect. of A span $\mathbb{R}^q$.

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PS (c) :

we again consider the map

$$\underline{x} \mapsto A\underline{x}.$$

(i) $\Rightarrow$ col vect span $\mathbb{R}^p \Rightarrow$ rows are lin. independent.

pf -

Since

$$A\underline{x} = x_1 \underline{c}_1 + \dots + x_p \underline{c}_p$$

col. vectors spanning $\mathbb{R}^p \Rightarrow \forall \underline{x} \in \mathbb{R}^p \vee \underline{b} \in \mathbb{R}^p, \exists \underline{x} \in \mathbb{R}^p :$

$$A\underline{x} = \underline{b}$$

This implies in particular that no row of A is all 0, otherwise if its the j^{th} row,

$$A\underline{x} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \rightarrow j$$

would have no solution.

But if the rows were lin ~~independent~~, i.e $\exists$

c_i not all zero such that:

$$c_1 \underline{r}_1^T + \dots + c_p \underline{r}_p^T = \underline{0}$$

, then if c_j is non-zero,

scale row $\underline{r}_j$ by c_j , and add $c_1 \underline{r}_1, c_2 \underline{r}_2, \dots, c_p \underline{r}_p$

for all indices not $= j$. This would produce a zero

row by our assumption, so that all $\underline{b} \in \mathbb{R}^p$

would not have a solⁿ, contradiction!

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(ii) Rows are lin. independent $\Rightarrow \underline{c}_1, \dots, \underline{c}_q$ span $\mathbb{R}^p$

Consider the contrapositive, suppose $\underline{c}_i$'s don't span $\mathbb{R}^p$,

i.e. $\exists \underline{b} : A\underline{x} = \underline{b}$ has no solution. Note that this must

imply that ~~some row of A is all zero~~ rows of A
are not linearly independent, because converting ~~$A \rightarrow \underline{b}$~~ $(A | \underline{b})$

to $(A_{RREF} | \underline{b}')$ we note that there must exist a solution
by back substitution, unless some row of A_{RREF} is

all zero. Since rows of A_{RREF} are just lin. combinations
of rows of A (as prev. discussed), and we cannot have

all ~~c_i~~ weights in the lin. combination be zero

(because we cannot scale a row by zero), these rows

must be lin. dependent! Thus we are done.